Expt 5 Design of High Pass Filter using Windowing Techniques

5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques

Objective:

1. To design an FIR High-Pass Filter using the Hamming Window technique for the given cutoff frequency, sampling frequency, and filter order specifications.

2. To generate the FIR filter coefficients using the Hamming Window method in MATLAB and implement the designed filter.

3. To obtain and analyze the magnitude response and phase response of the FIR High-Pass Filter designed using the Hamming Window.

Procedure:

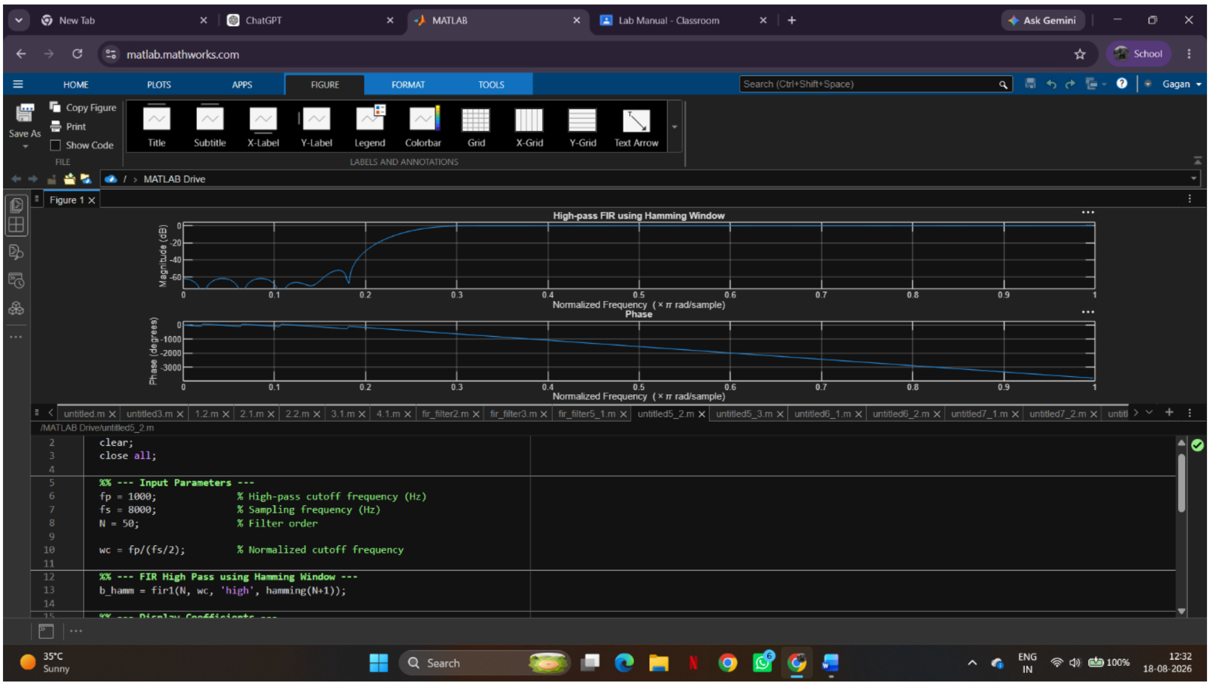
1. Open MATLAB and create a new script file.
2. Clear the workspace, command window, and close all figure windows.
3. Specify the sampling frequency, cutoff frequency, and filter order.
4. Calculate the normalized cutoff frequency.
5. Design the FIR High-Pass Filter using the fir1() function with the 'high' option and a Hamming Window (hamming).
6. Generate and display the FIR filter coefficients.
7. Obtain the magnitude response and phase response of the designed filter using the freqz() function.
8. Plot and analyze the frequency response characteristics of the filter.
9. Verify the high-pass filtering behavior by observing the attenuation of low-frequency components and the passage of high-frequency components.
10. Record the observations and conclude the performance of the FIR High-Pass Filter designed using the Hamming Window technique.

Program (Matlab Code):

```
clc;

clear;

close all;
```

```
%% --- Input Parameters ---
fp = 1000;            % High-pass cutoff frequency (Hz)
fs = 8000;            % Sampling frequency (Hz)
N = 50;               % Filter order

wc = fp/(fs/2);       % Normalized cutoff frequency

%% --- FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));

%% --- Display Coefficients ---
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm');

%% --- Frequency Response Plot ---
figure;
freqz(b_hamm, 1);
title('High-pass FIR using Hamming Window');
grid on;
```

OUTPUT:



Result:

1. The FIR High-Pass Filter was successfully designed using the Hamming Window technique for the specified cutoff frequency, sampling frequency, and filter order.

2. The FIR filter coefficients were successfully generated using the Hamming Window method in MATLAB and the filter was implemented.

3. The magnitude response and phase response of the designed FIR High-Pass Filter were obtained and analyzed. The magnitude response showed effective attenuation of low-frequency components and passage of high-frequency components, confirming the desired high-pass filtering behavior. The phase response exhibited the expected linear-phase characteristics of an FIR filter.